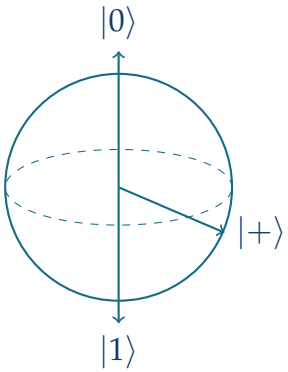
QUANTUM TECHNOLOGIES

Lab Assignment

Execution of Six Algorithms using Simulation and IBM Quantum Environment

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1 Deutsch Algorithm

1.1 Simulated Output

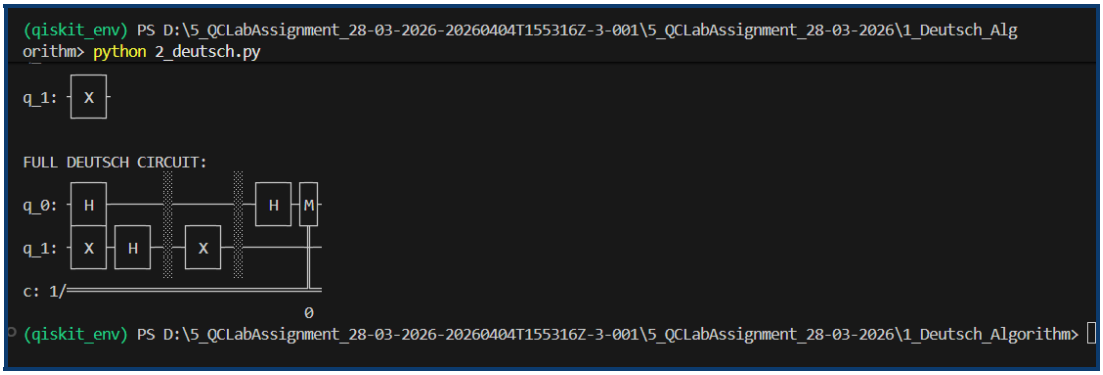


Figure 1: Output of Deutsch Algorithm on Local Simulation

1.2 IBM Quantum Environment Output

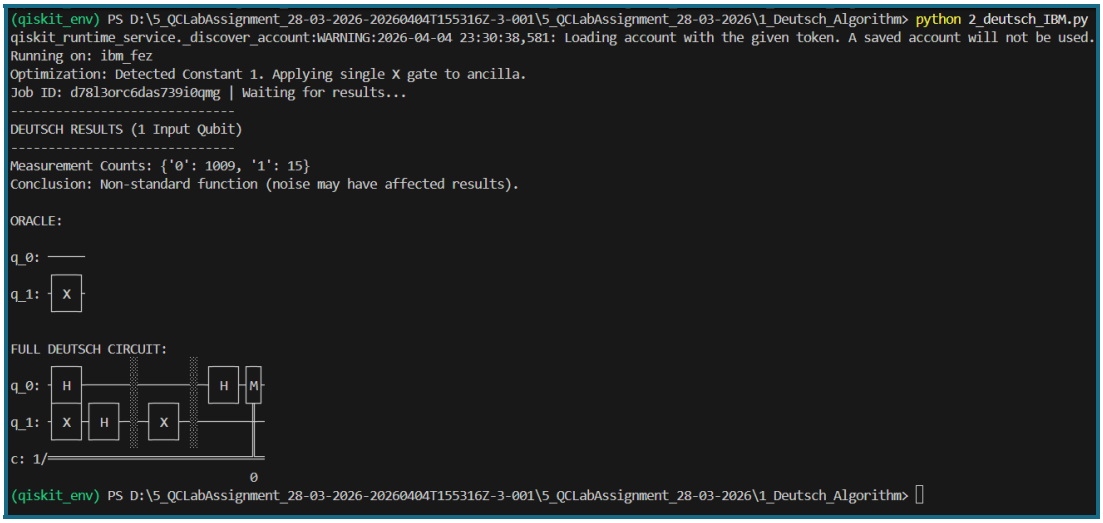


Figure 2: Output of Deutsch Algorithm on IBM Quantum Environment

2 Deutsch-Jozsa Algorithm

2.1 Simulated Output

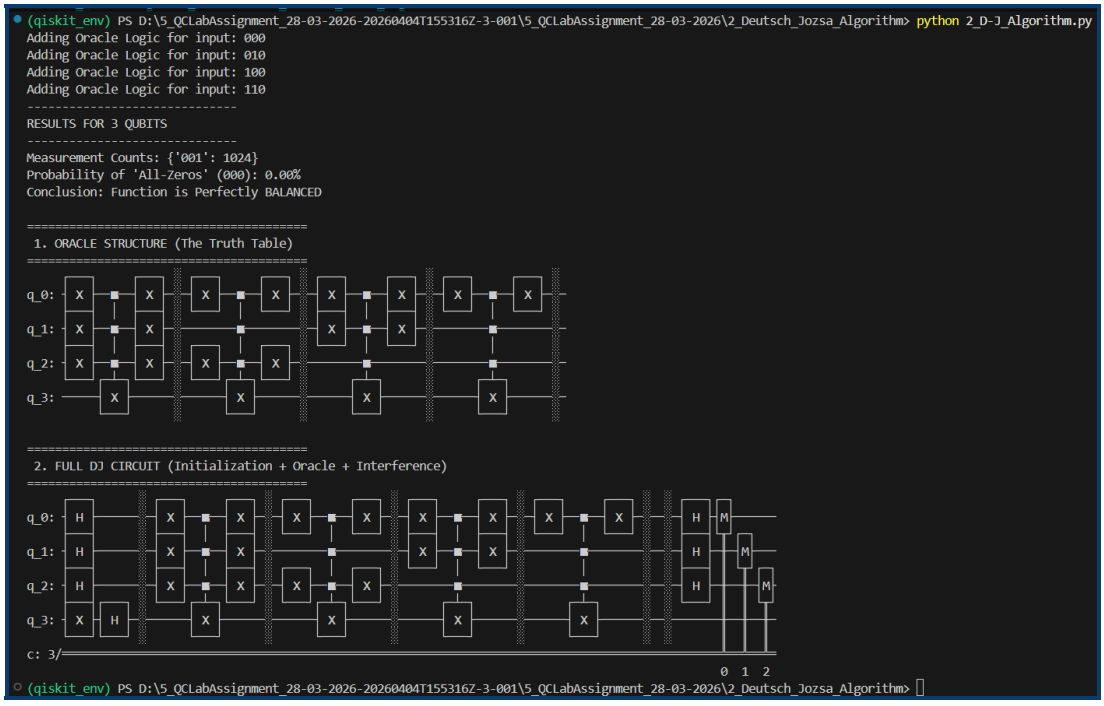


Figure 3: Output of Deutsch-Jozsa Algorithm on Local Simulation

2.2 IBM Quantum Environment Output

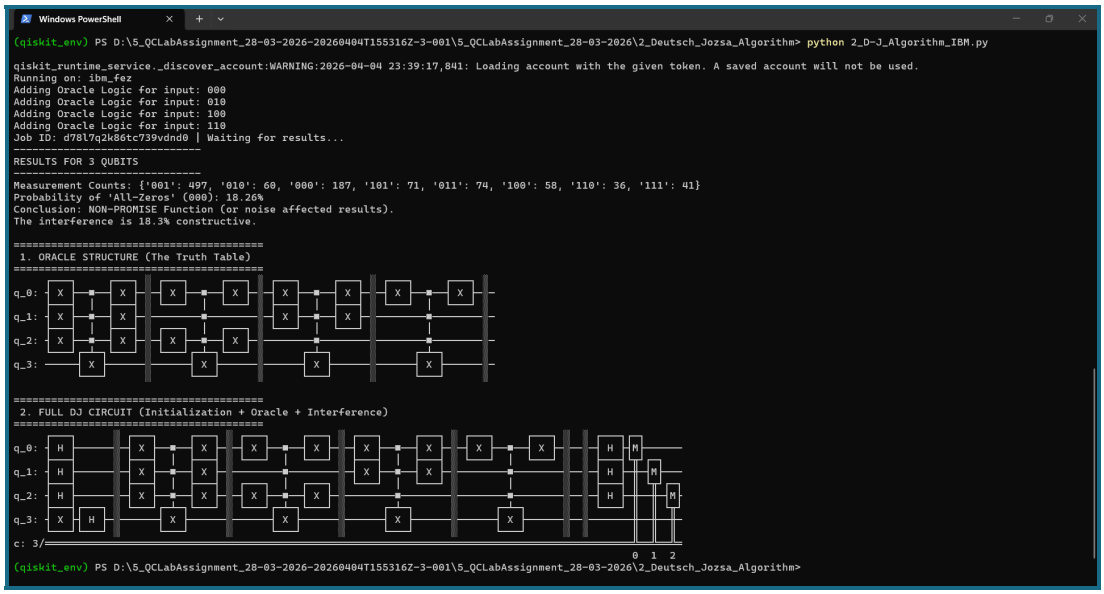


Figure 4: Output of Deutsch-Jozsa Algorithm on IBM Quantum Environment

3 Bernstein-Vazirani Algorithm

3.1 Simulated Output

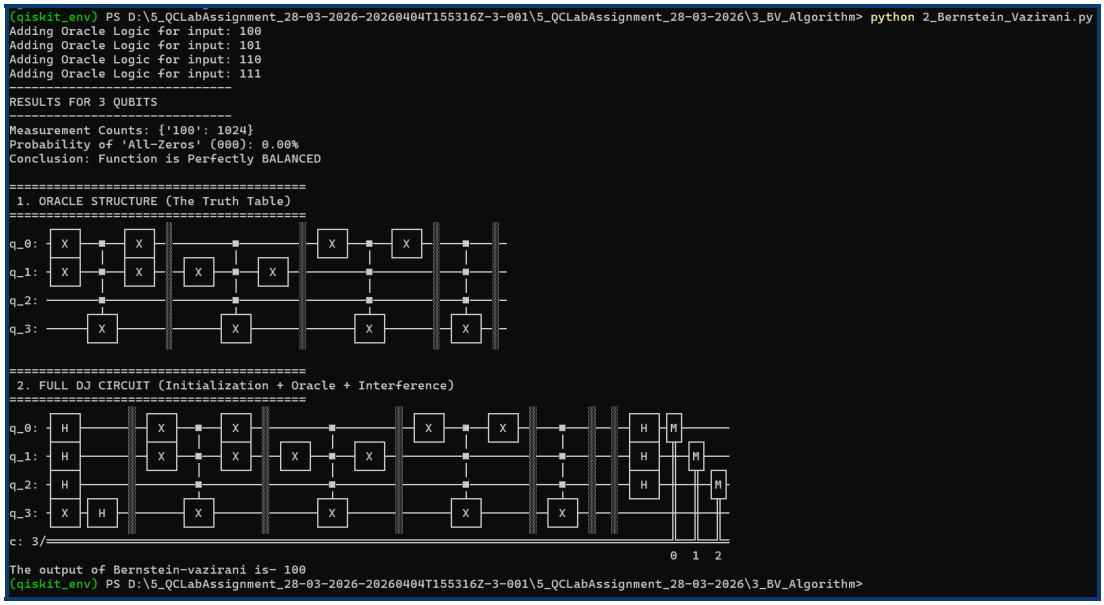


Figure 5: Output of Bernstein-Vazirani Algorithm on Local Simulation

3.2 IBM Quantum Environment Output

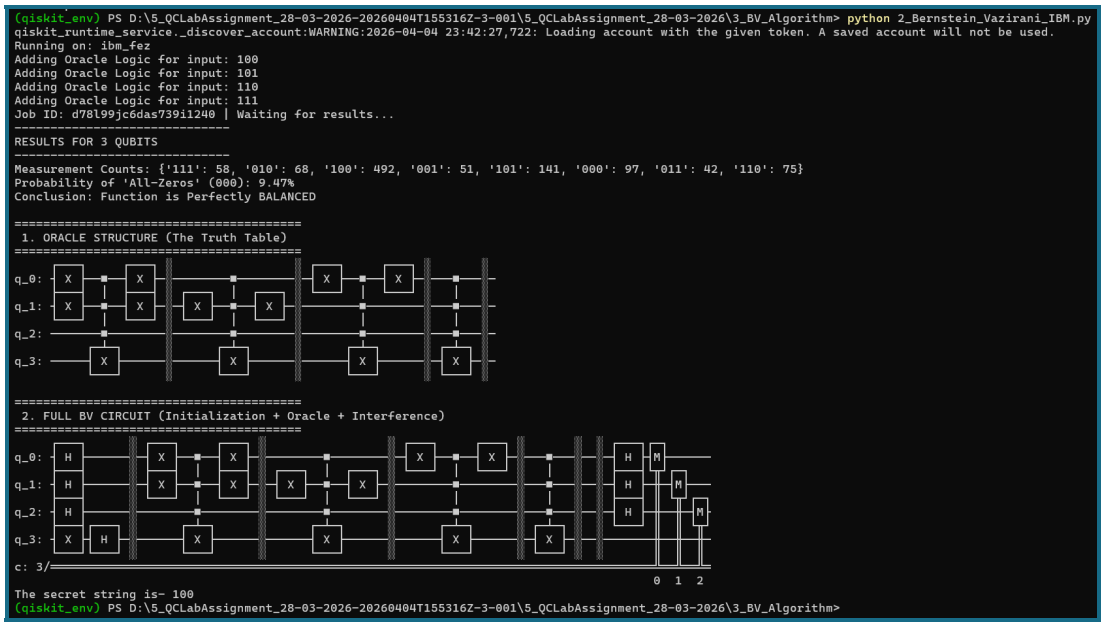


Figure 6: Output of Bernstein-Vazirani Algorithm on IBM Quantum Environment

4 Simon's Periodicity Algorithm

4.1 Simulated Output

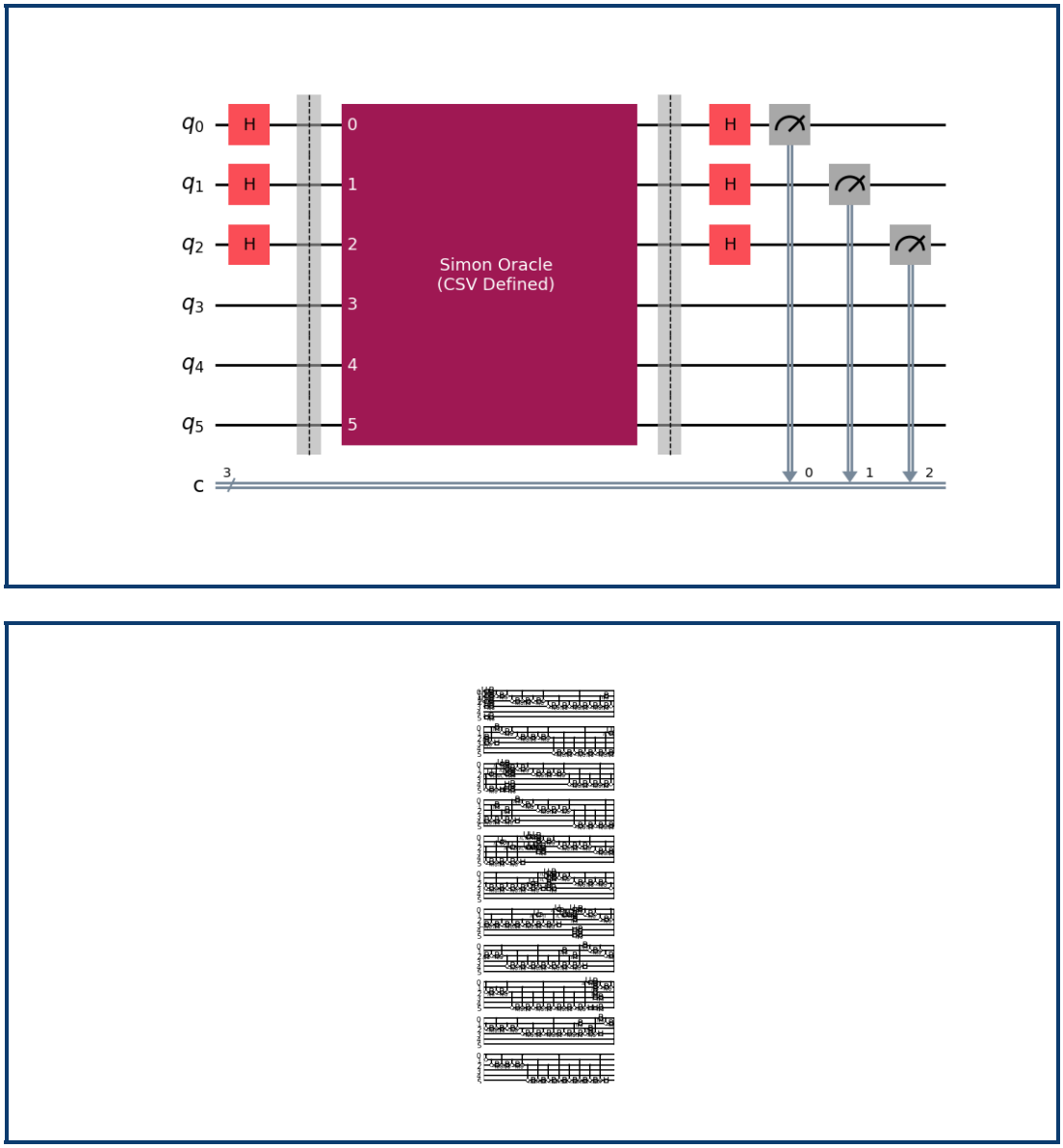
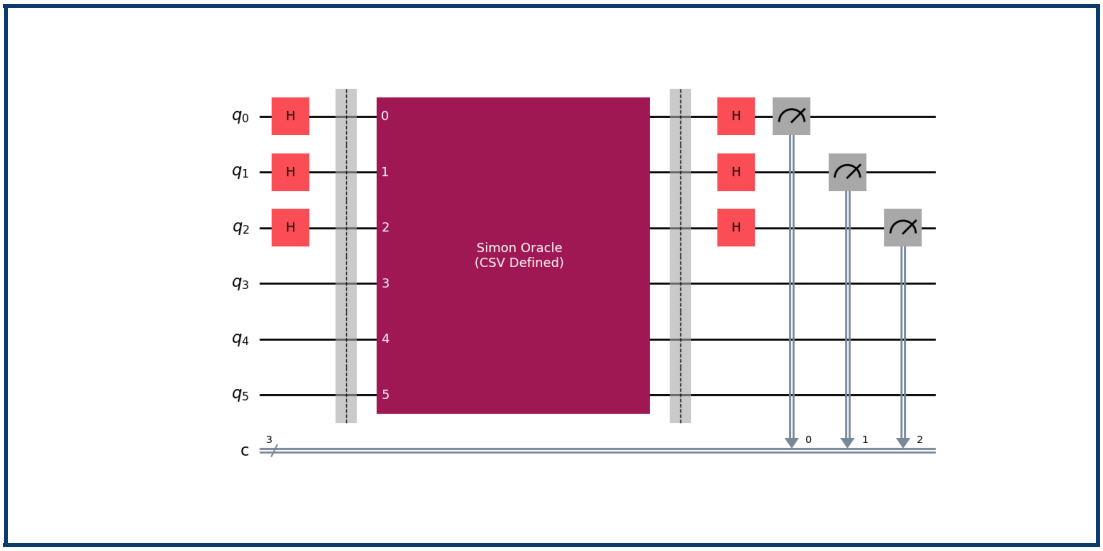


Figure 7: Output of Simon's Periodicity Algorithm on Local Simulation

4.2 IBM Quantum Environment Output



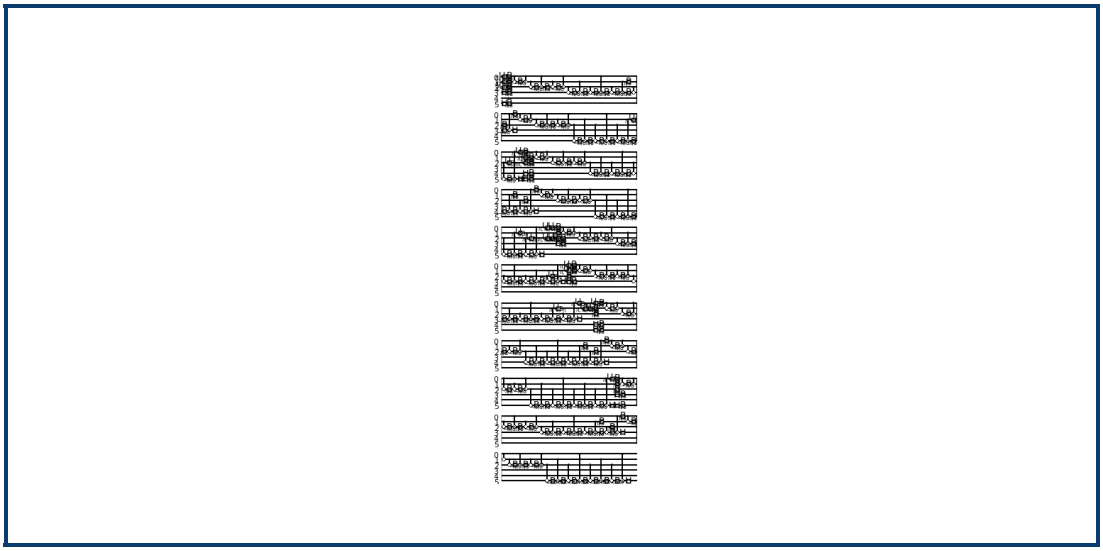


Figure 8: Output of Simon's Periodicity Algorithm on IBM Quantum Environment

4.3 Eq Solving Classical

```
(qiskit_env) PS D:\5_QCLabAssignment_28-03-2026-20260404T155316Z-3-001\5_QCLabAssignment_28-03-2026\4_Simons_Algorithm> python 3_eq_solving_classical.py
Solving for n=3...
Secret string is 000 (or insufficient equations).
(qiskit_env) PS D:\5_QCLabAssignment_28-03-2026-20260404T155316Z-3-001\5_QCLabAssignment_28-03-2026\4_Simons_Algorithm>
```

Figure 9: Output of eq solving python file

5 Grover's (Search) Algorithm

5.1 Simulated Output

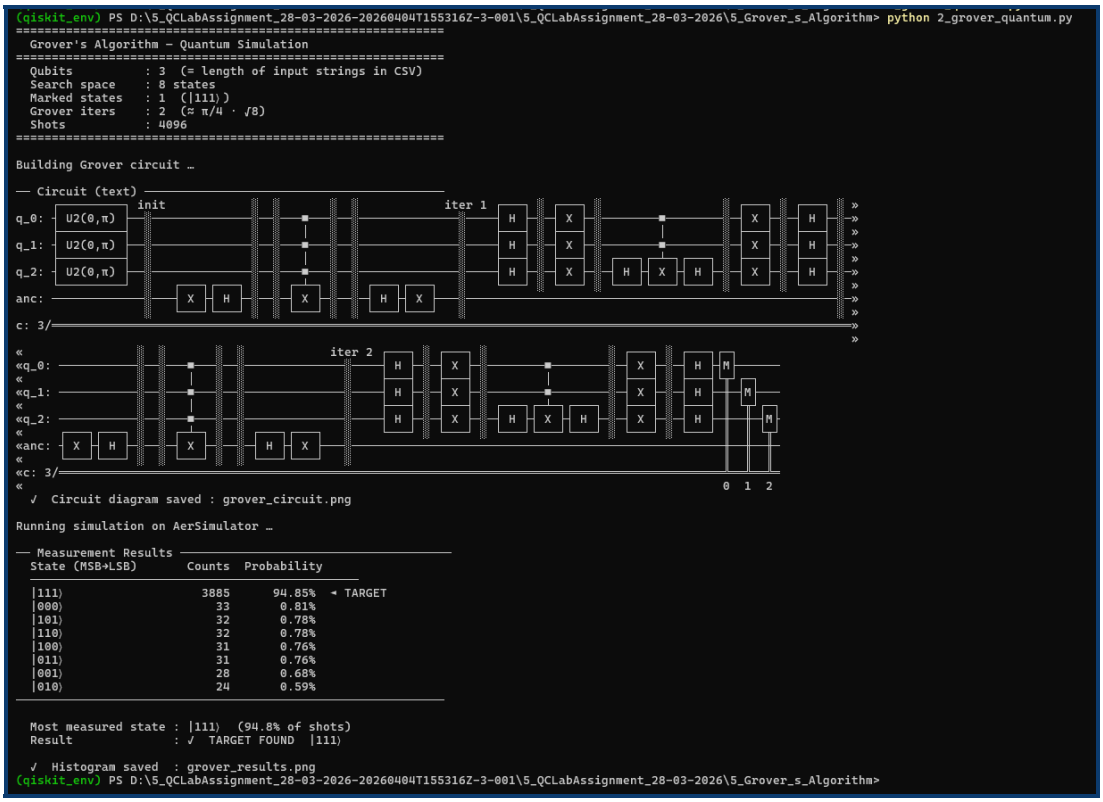


Figure 10: Output of Grover's Algorithm on Local Simulation

5.2 Simulated Output Histogram

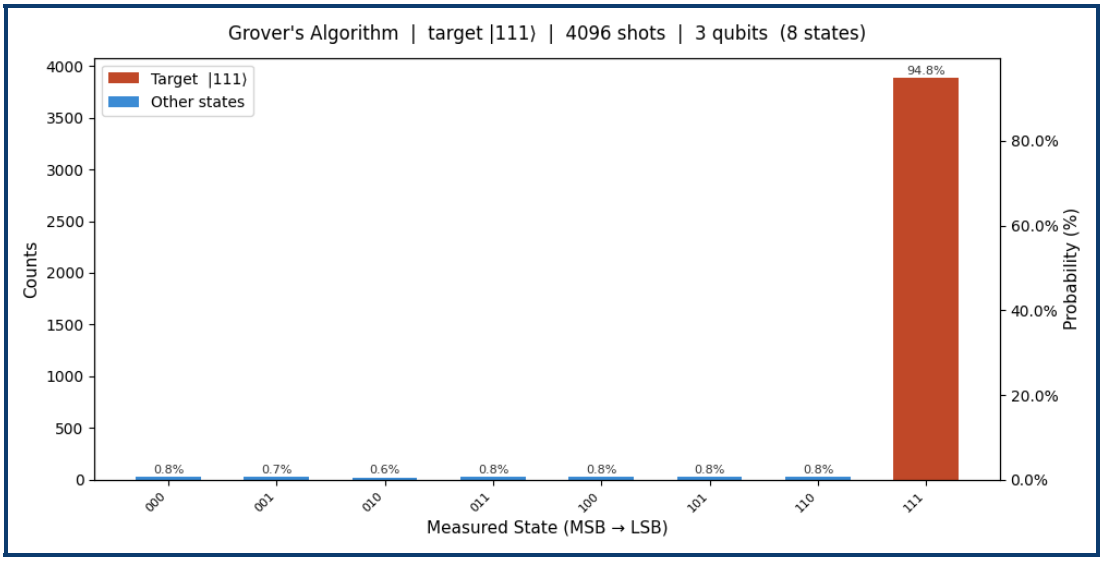


Figure 11: Histogram Output of Grover's Algorithm on Local Simulation

5.3 IBM Quantum Environment Output

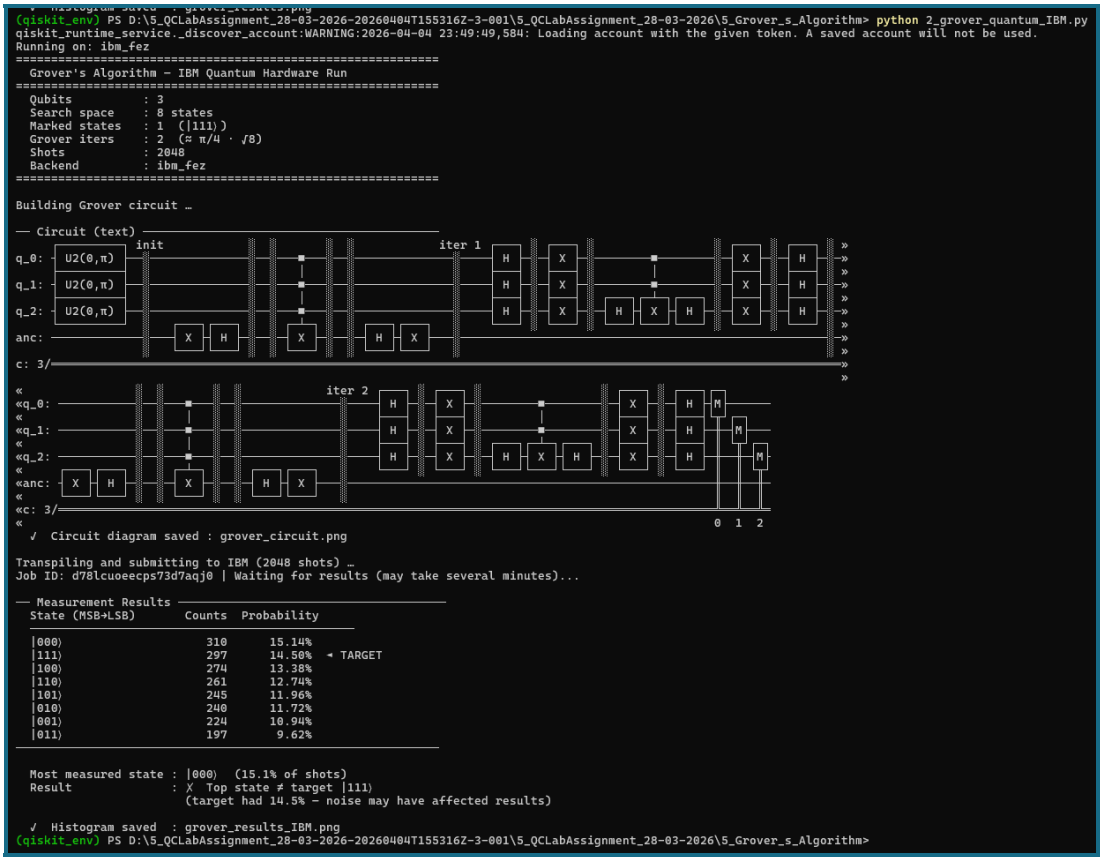


Figure 12: Output of Grover's Algorithm on IBM Quantum Environment

5.4 IBM Quantum Environment Output Histogram

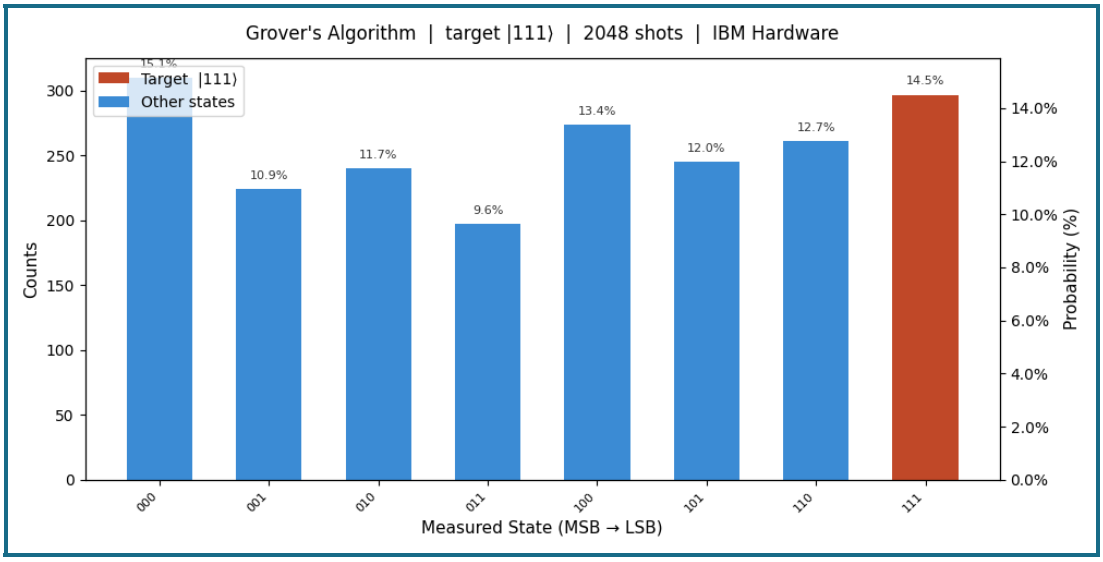


Figure 13: Histogram Output of Grover's Algorithm on IBM Quantum Environment

6 Shor's (Prime Factors) Algorithm

6.1 Generation Output

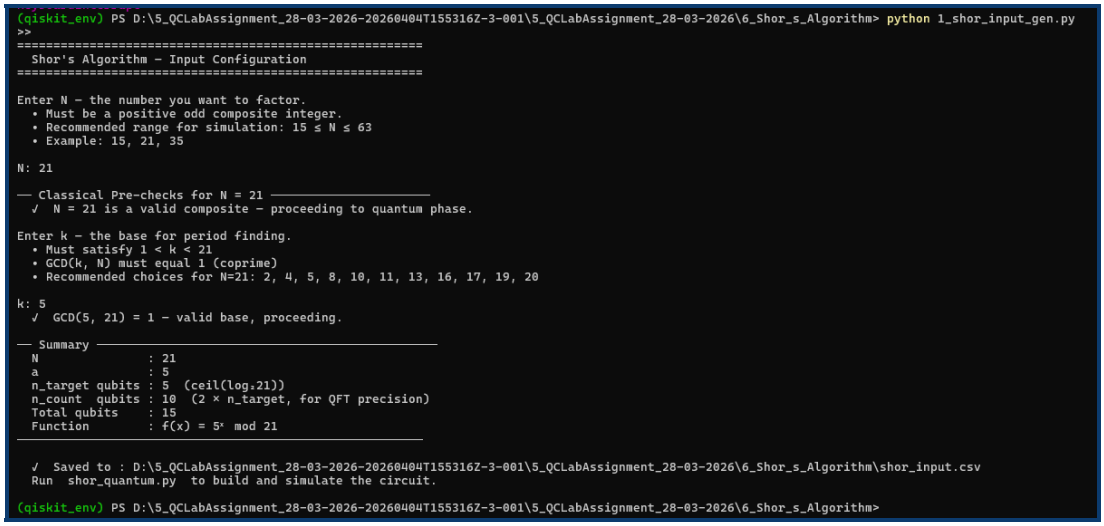


Figure 14: Screenshot Of Shor's input gen file

6.2 Simulated Output

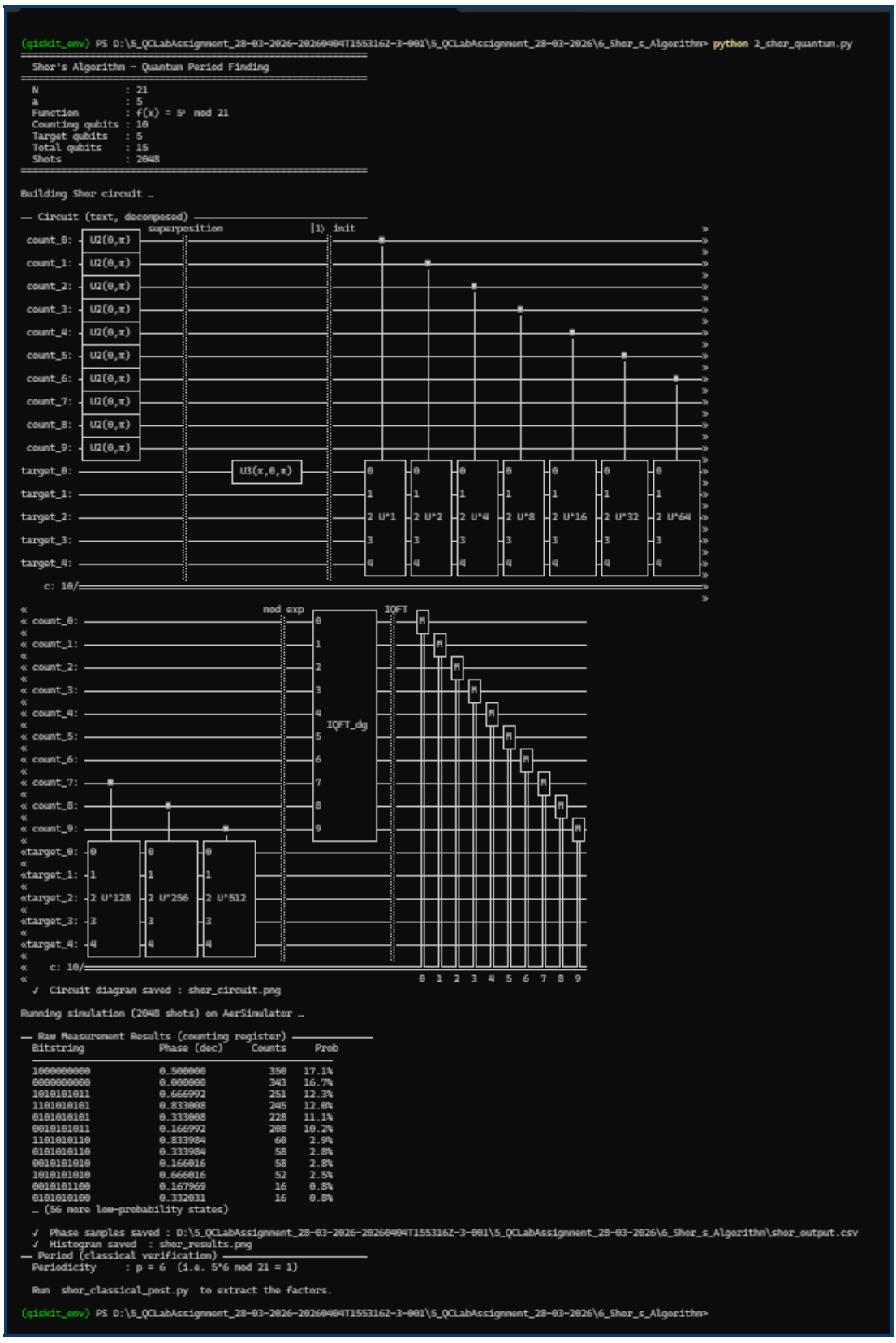


Figure 15: Output of Shor's Algorithm on Local Simulation

6.3 Simulated Output Histogram

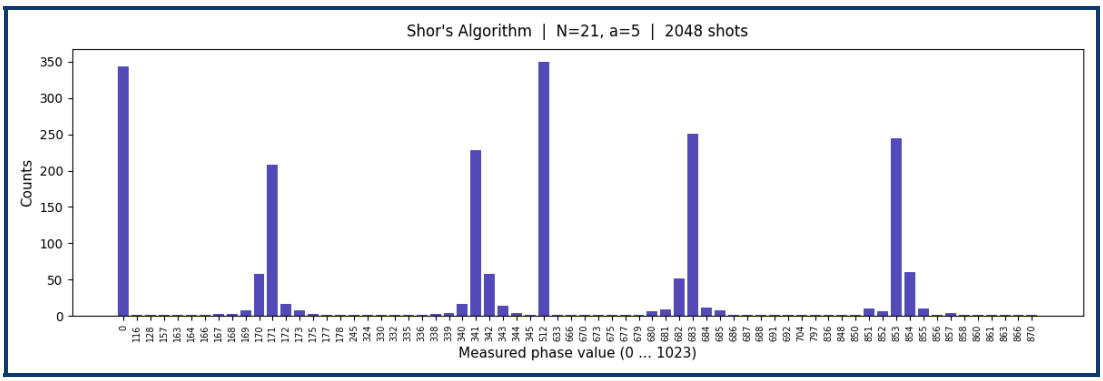


Figure 16: Histogram Output of Shor's Algorithm on Local Simulation

6.4 IBM Quantum Environment Output

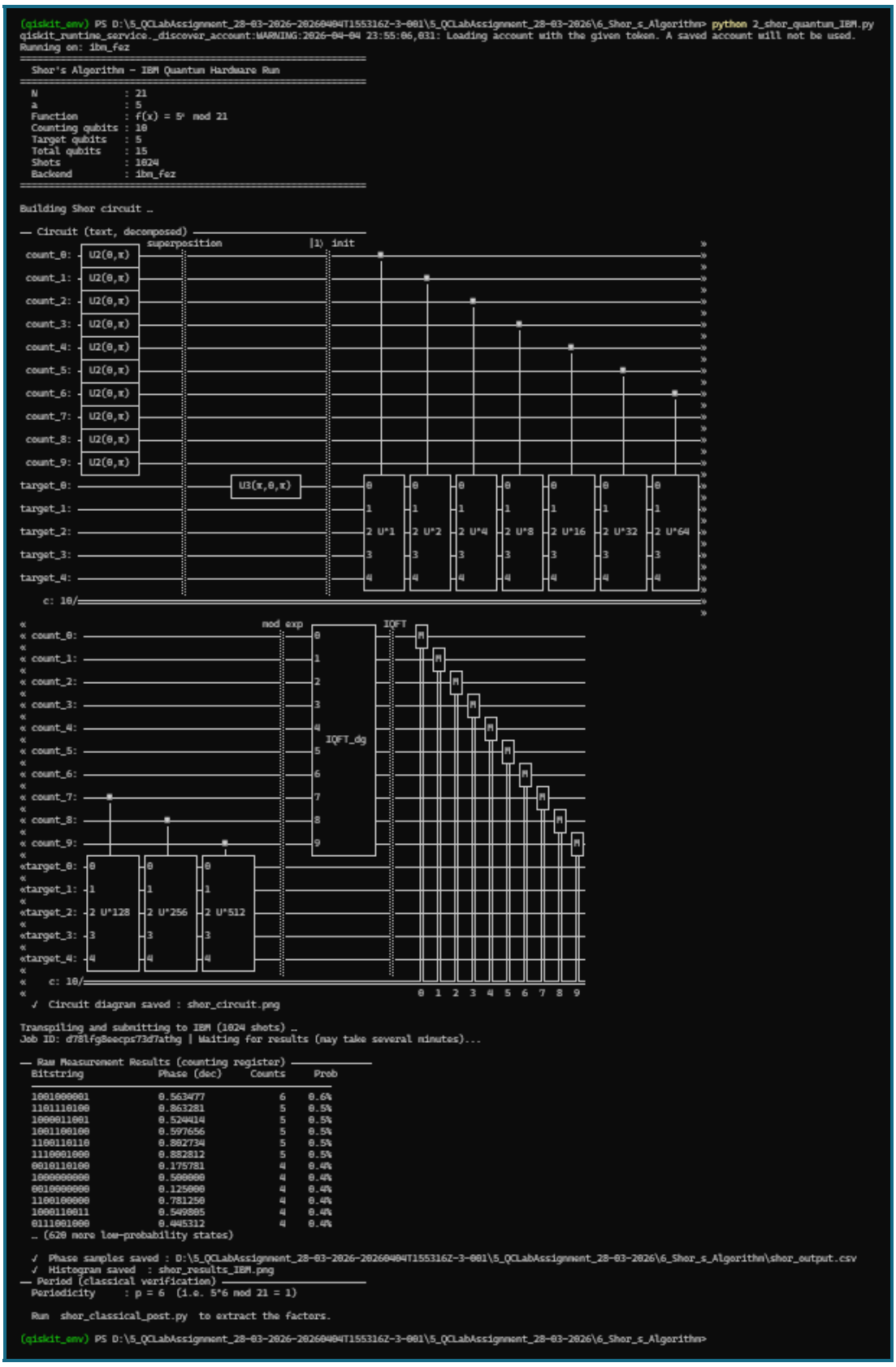


Figure 17: Output of Shor's Algorithm on IBM Quantum Environment

6.5 IBM Quantum Environment Output Histogram

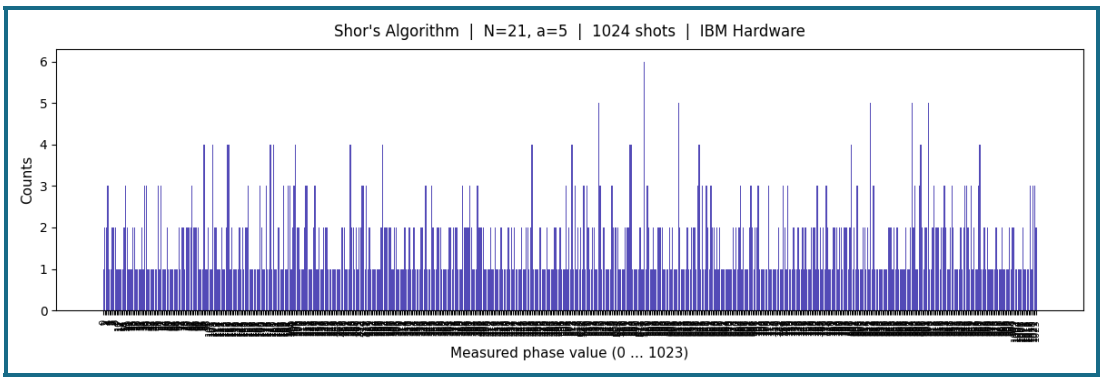


Figure 18: Histogram Output of Shor's Algorithm on IBM Quantum Environment